

Aaryan Lath

Systems and Design Engineer | (312) 838-5669 | aaryanlath05@gmail.com
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EDUCATION

Purdue University

Bachelor of Science in Aeronautical and Astronautical Engineering (GPA: 3.74/4.0)

Master of Science in Aeronautics and Astronautics (GPA: 4.0/4.0)

Specialization in Aerodynamics and Systems Design

Honors: Dean's List and Semester Honors for **every** semester, **AAE Ambassador**

West Lafayette, IN

Expected May 2026

Expected May 2027

SKILLS

Core Tools: CAD, Assembly, CREO, Siemens NX, GD&T, Teamcenter, Ansys, Mechanical Design, JIRA, SAP

Programming & Analysis: Technical Writing, Python, MATLAB, Trade Studies, Git, CFD, Simulink, FEA

Soft-skills: Analytical, Communicator, Detail-oriented, Fast-paced learner, Inquisitive

Awards: 2nd place in Nationals of World Robot Olympiad Junior High

PROFESSIONAL EXPERIENCE

Zucrow Laboratories (High Speed Compressors Lab)

Undergraduate Research Assistant

West Lafayette, IN

Aug 2025 – Present

- Researching into the inlet vortex distortion phenomena and their effects on turbomachinery performance.
- Helped assemble and calibrate the test cell's fan-rig using laser alignment, reducing time taken by **15%**.

Siemens USA

Mechanical Engineering Intern

Grand Prairie, TX

May 2025 – Aug 2025

- Designed custom enclosures for panelboards using **CREO** and executed ECNs in **SAP**.
- Streamlined **100+** switchboard neutral assemblies configurations by resolving edge case designs.
- Automated BOM generation with Python, cutting manual processing time by **~30%** for **10+** orders/day.

School of Aeronautics and Astronautics

Undergraduate Teaching Assistant

West Lafayette, IN

Jan 2025 – Present

- Led study sessions to teach **50+** students, core **AAE 251** (Aircraft and Spacecraft Design) course material.
- Provided personalized support to students with concepts and **MATLAB** troubleshooting.
- Guided students in understanding key design principles for the aircraft and spacecraft design project.

Resilient Extraterrestrial Habitat Engineering

Undergraduate Research Assistant - Systems Engineer

West Lafayette, IN

May 2024 – May 2025

- Analyzed **18** potential disruptions scenarios in a lunar habitat through conducting **Simulink** simulations.
- Performed trade studies to evaluate safety controls, optimizing **resilience** and costs using **ESM** analysis.
- Executed **FEA** simulations and tests to validate **vibration isolation** devices under various conditions.

PROJECTS AND INVOLVEMENTS

SAE Aero Design

Chief Engineer/ Program Manager

West Lafayette, IN

May 2025 – Present

- Authored a **~75 page** RC aircraft design guide to accelerate onboarding
- Documented past failures from the SAE Aero Design Regular Class to avoid recurring errors.
- Led technical development for **3** sub teams by reviewing trade studies and ensured milestones are met.

Purdue Space Program – A SEDS Chapter

Technical Director

West Lafayette, IN

Dec 2024 – May 2025

- Optimized the clubs backend operations and handled 9 sub-teams for smooth project completion.

Satellites Deputy Systems Director

Feb 2024 – Nov 2024

- Improved project progress with **~15** engineers and other sub-teams ensuring alignment with project goals.
- Formulated, reviewed, and refined **200** project and system-level requirements for Boiler Bus.
- **Manufactured** the satellite frame end plates using **CAM** and walls through **laser cutting**.

Single Piston Sterling Engine

Test Engineer

West Lafayette, IN

Oct 2024 – Jan 2025

- Tested for end-to-end connectivity between **NX** and **Teamcenter** for students in preparation for deployment.
- Reported bugs in Engineering Change Orders (**ECOs**) and Product Data Management (**PDM**) workflow.